(iii) The potential difference (p.d.) across the motor has a constant value of 600 V and the motor has an efficiency of 85%.

Calculate the current in the motor.

current = ... A [3]

(c) The car in **(b)** now reaches a slope, as shown in Fig. 3.2.

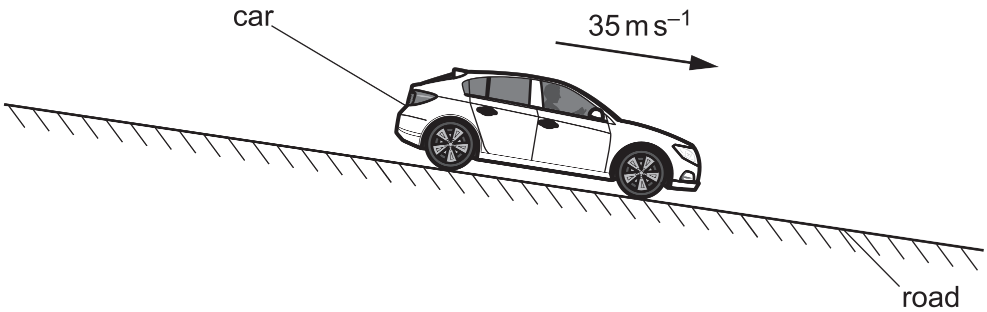

Fig. 3.2

The car continues down the slope at the same speed as in **(b)**.

State and explain the effect, if any, of the slope on:

(i) the air resistance acting on the car

....

.... [1]

(ii) the current in the motor.

....

.... [1]

[Total: 10]